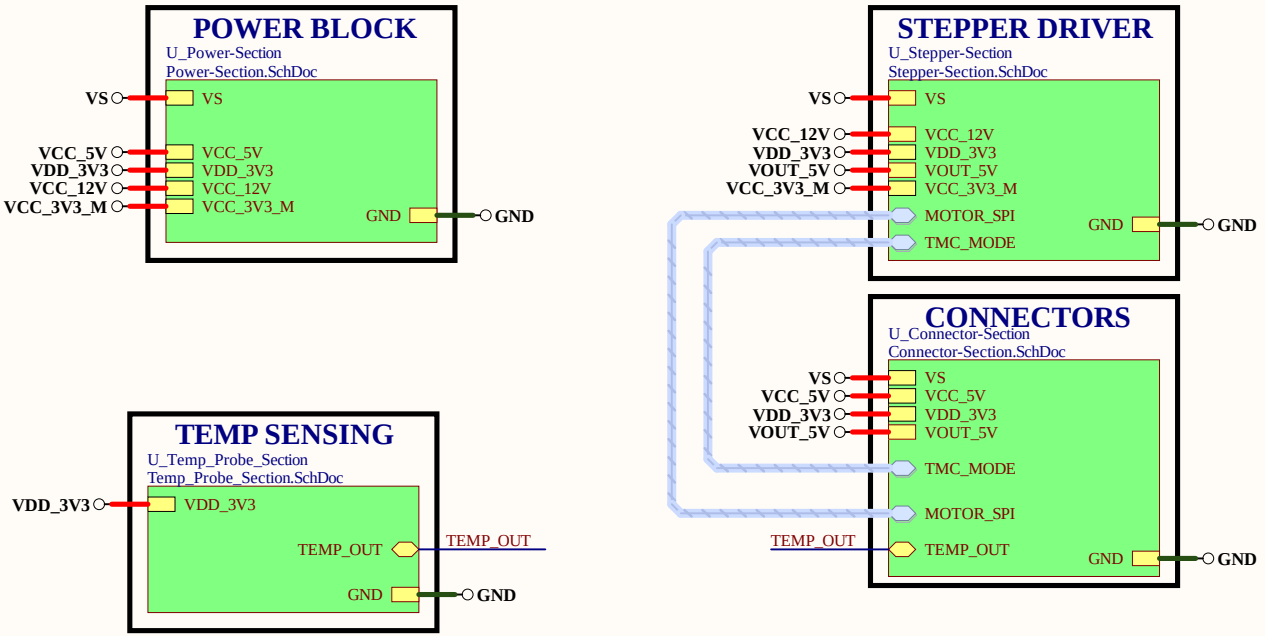
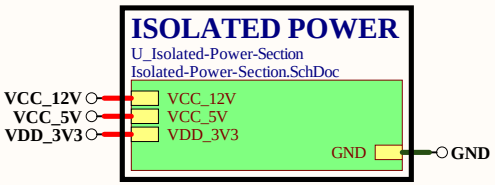
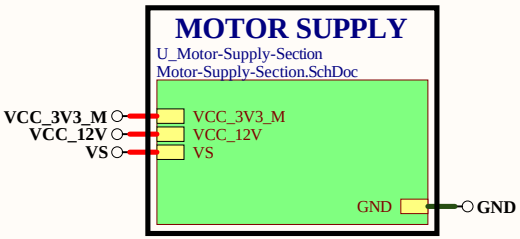
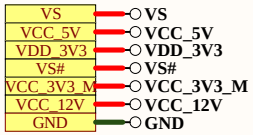
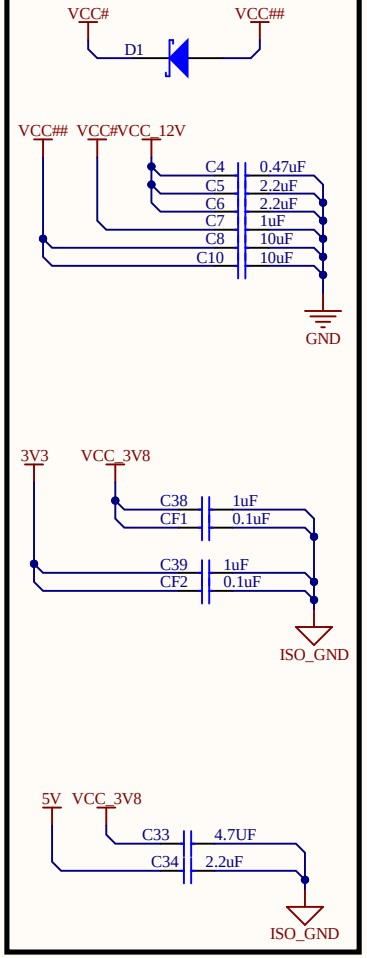

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
VS
VCC_12V
VDD_3V3
VOUT_5V
VCC_3V3_M
MOTOR_SPI
TMC_MODE
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VS
VCC_5V
VDD_3V3
VOUT_5V
TMC_MODE
MOTOR_SPI
TEMP_OUT
GND
GND
TEMP SENSING
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
TEMP_OUT
GND
GND
TEMP_OUT


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
GND
VCC_12V
VCC_5V
VDD_3V3
GND

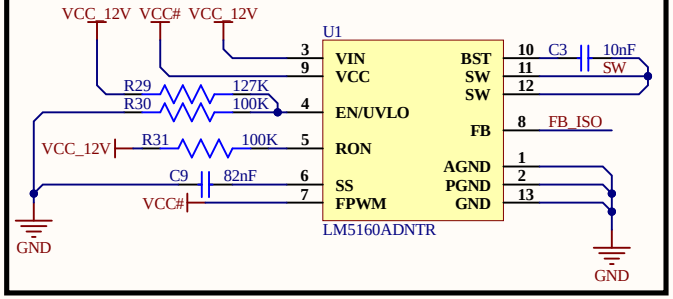

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VCC_3V3_M
VCC_12V
VS
VCC_3V3_M
VCC_12V
VS
GND
GND


VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND
VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND

## DECOUPLING CAPS

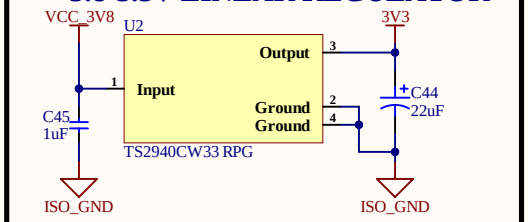



## 12V-3.8V ISOLATED BUCK CONVERTER

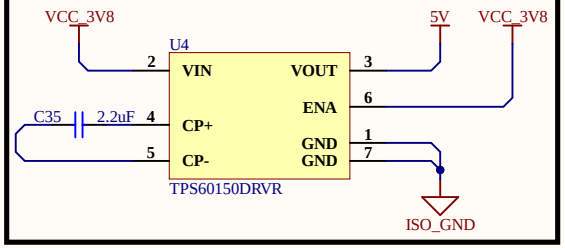



Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

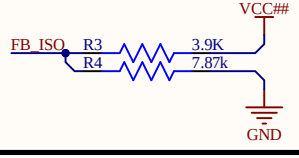
## 3.8-3.3V LINEAR REGULATOR



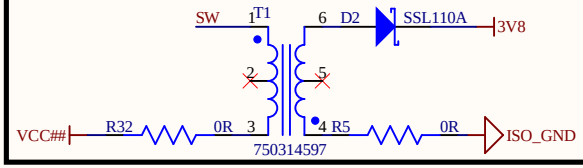

## 3.8-5V CHARGE PUMP



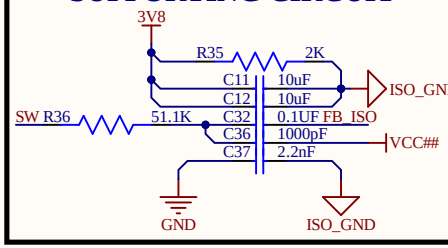

## FEEDBACK NETWORK



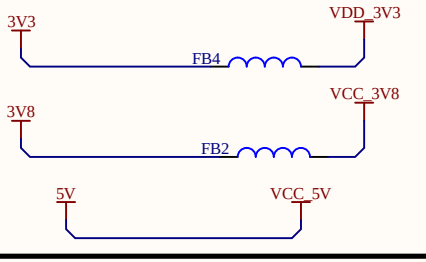

## FLYBACK TRANSFORMER




## SUPPORTING CIRCUIT

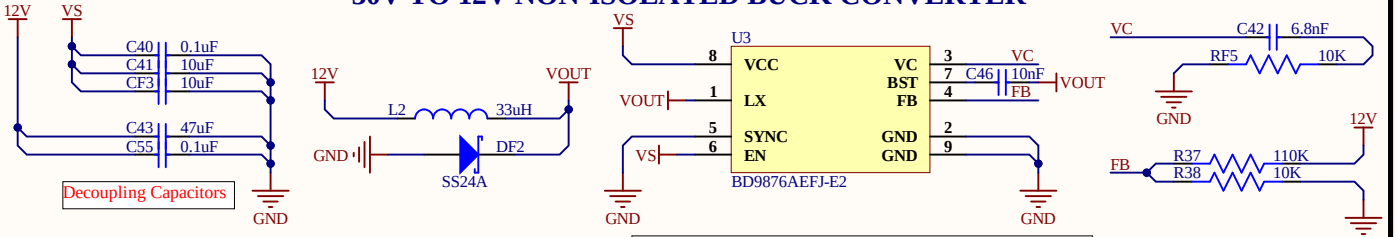



## FERRITE CONNECTIONS

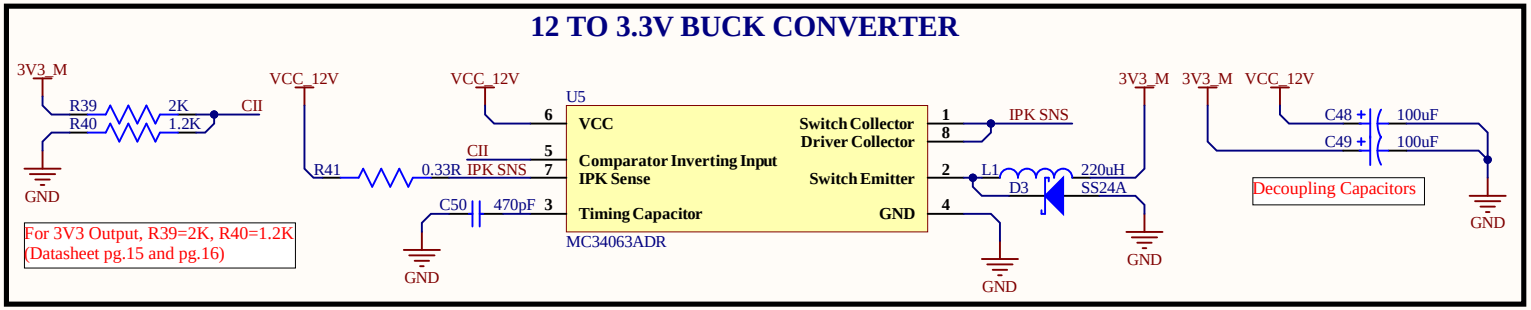



VCC_12V — VCC_12V
VCC_5V — VCC_5V
VDD_3V3 — VDD_3V3
GND — GND

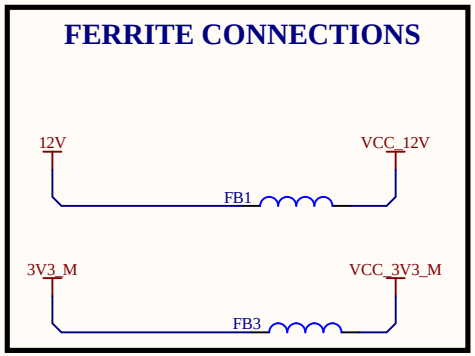
## 30V TO 12V NON-ISOLATED BUCK CONVERTER

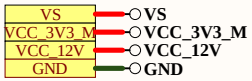


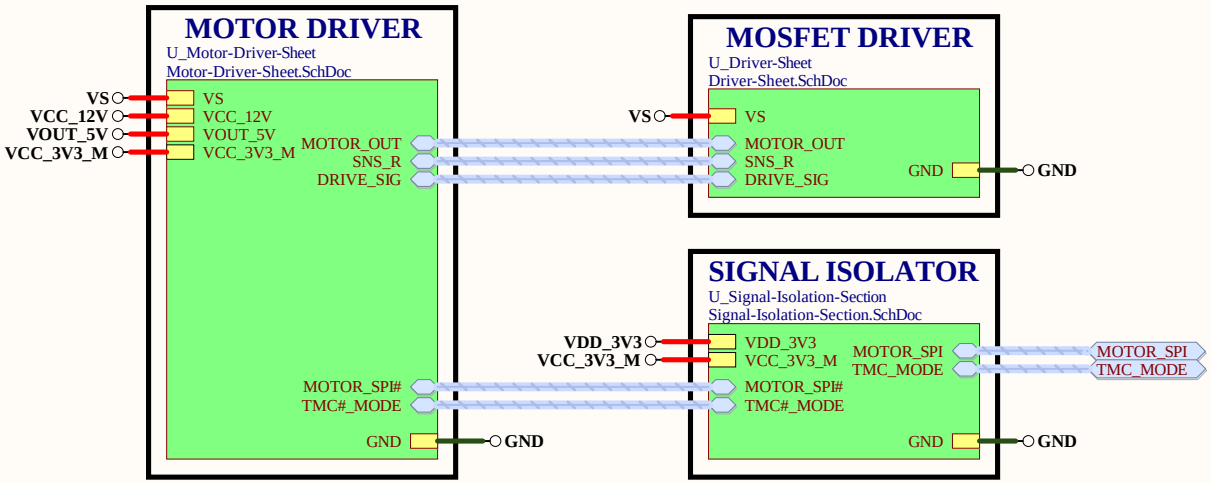

## 12 TO 3.3V BUCK CONVERTER

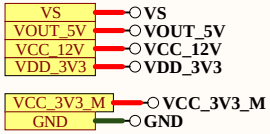


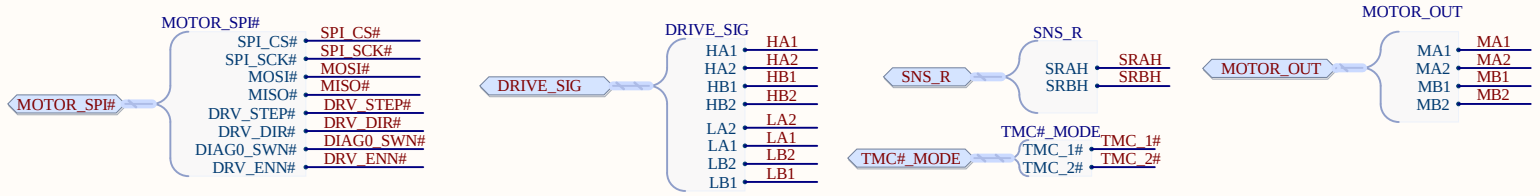

## FERRITE CONNECTIONS



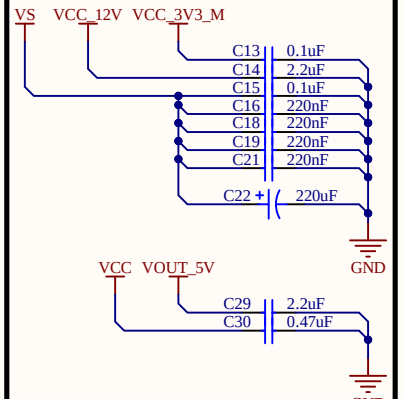




MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
MOTOR_SPI
TMC_MODE

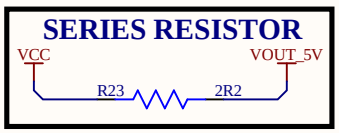

VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND



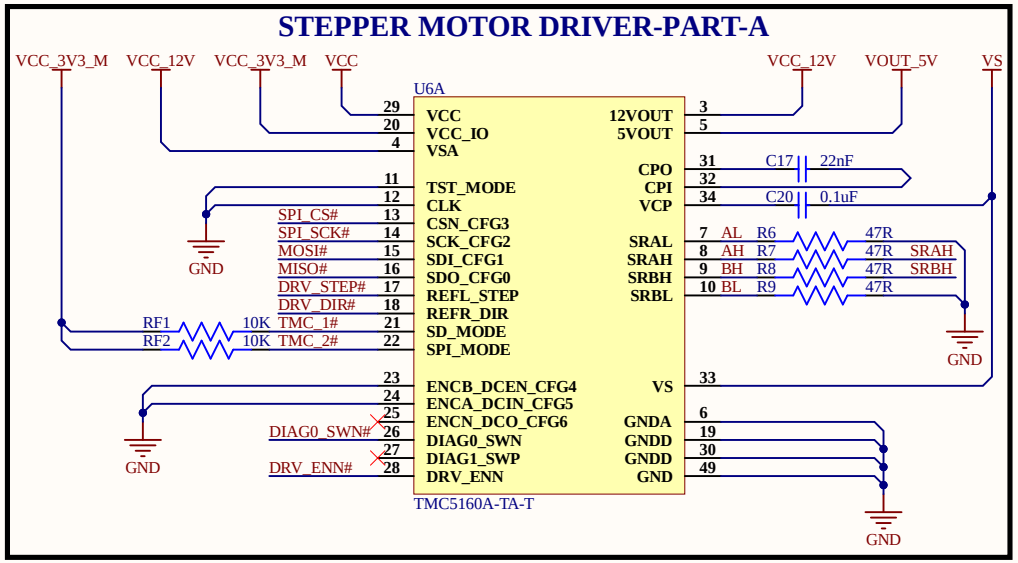

## DECOUPLING CAPS



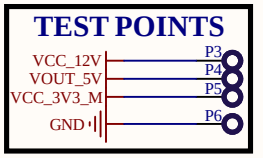

## SERIES RESISTOR



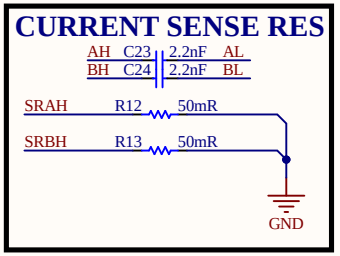

## STEPPER MOTOR DRIVER-PART-A



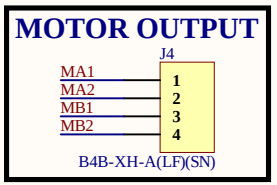

## TEST POINTS



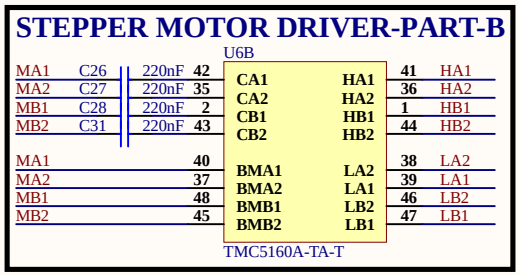

## CURRENT SENSE RES



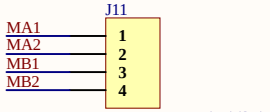

## MOTOR OUTPUT

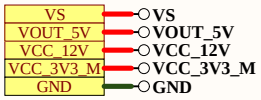


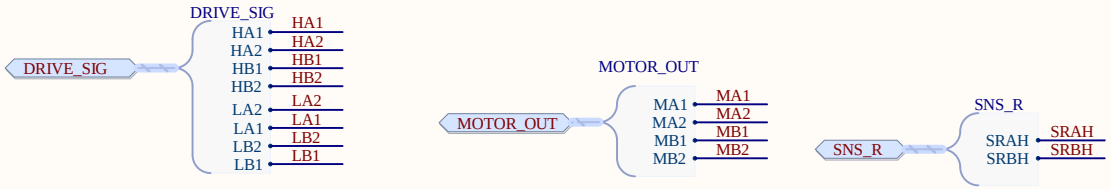

## STEPPER MOTOR DRIVER-PART-B



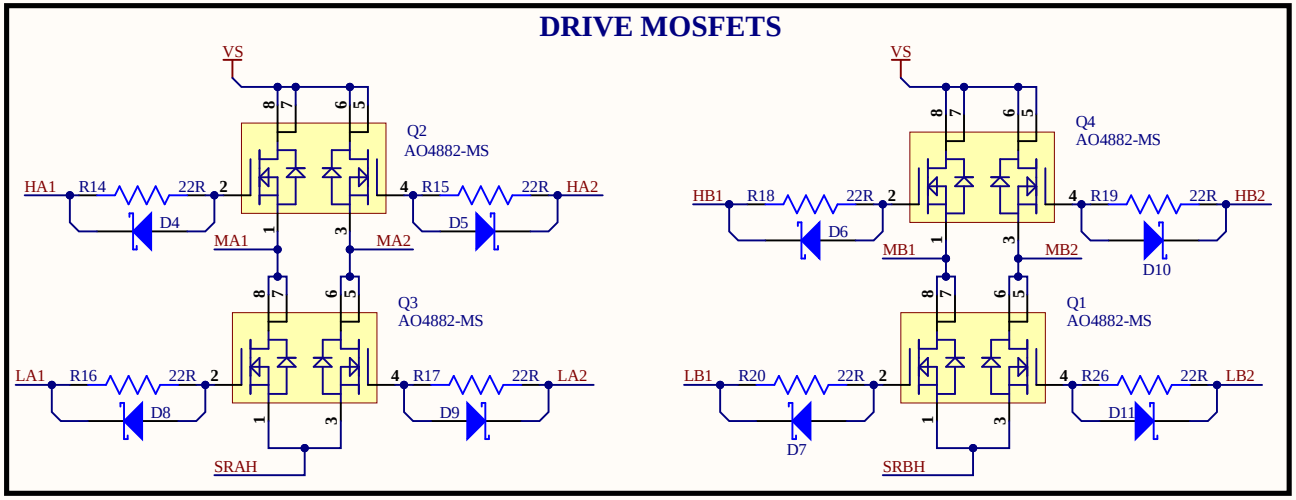

## MOTOR OUTPUT FOR NEMA34



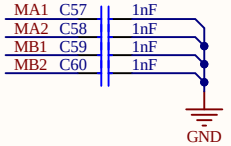





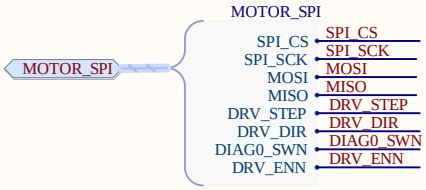

## DRIVE MOSFETS

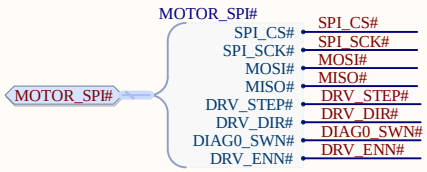



## DECOUPLING CAPS





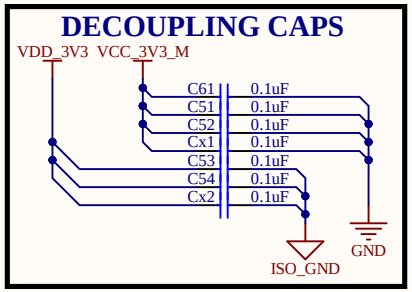



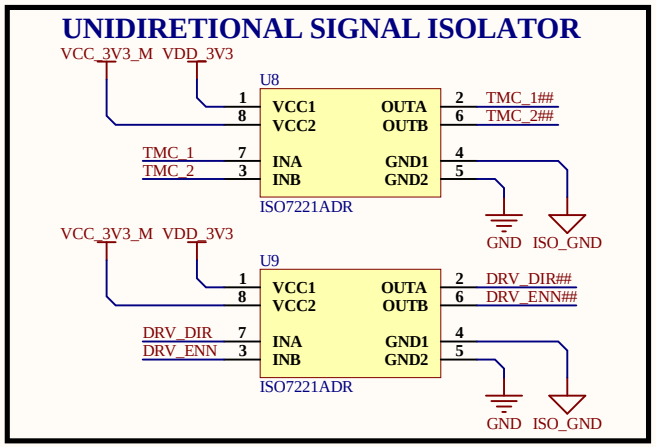



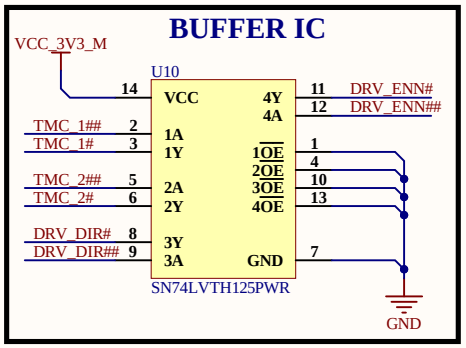


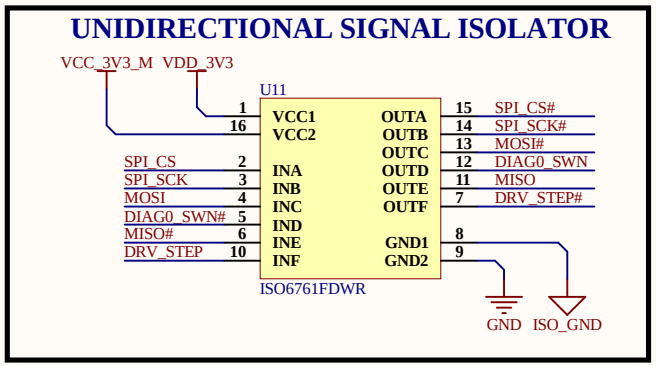




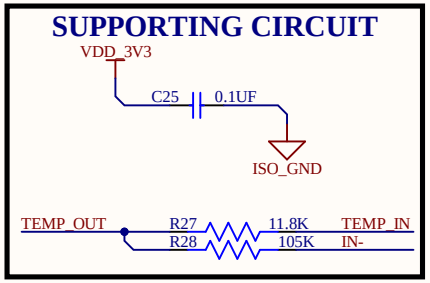




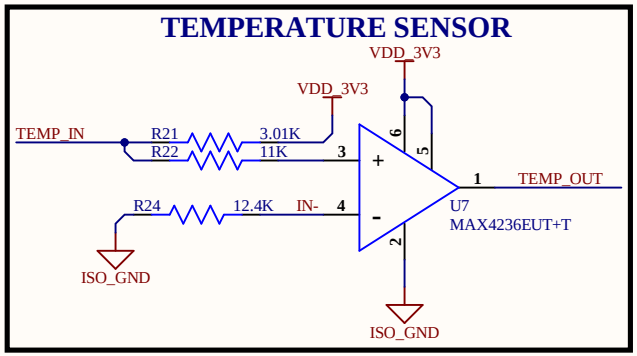
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| Date: 11/24/2023 | | Sheet of |
| File: C:\Users\..\Signal-Isolation-Section.SchDoc | | Drawn By: |

TEMP_OUT TEMP_OUT

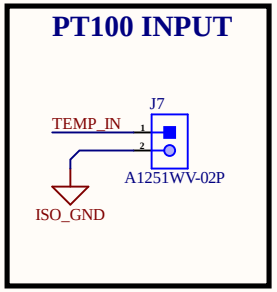
## SUPPORTING CIRCUIT



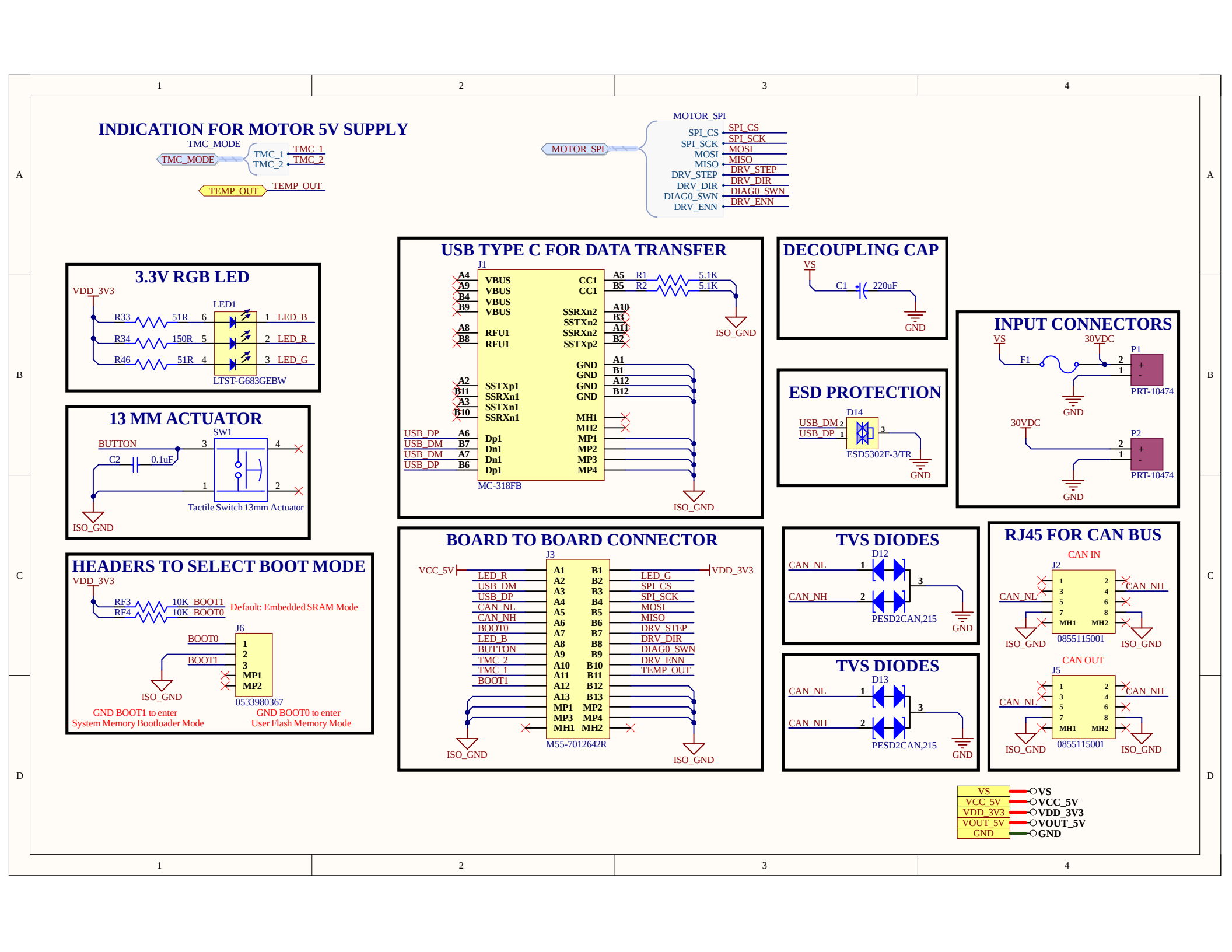

## TEMPERATURE SENSOR




## PT100 INPUT




VDD_3V3 VDD_3V3
GND GND


INDICATION FOR MOTOR 5V SUPPLY
TMC_MODE
TMC_1
TMC_2
TEMP_OUT
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
3.3V RGB LED
VDD_3V3
LED1
R33 51R
R34 150R
R46 51R
LED_B
LED_R
LED_G
LTST-G683GEBW
13 MM ACTUATOR
SW1
BUTTON
C2 0.1uF
Tactile Switch 13mm Actuator
ISO_GND
HEADERS TO SELECT BOOT MODE
RF3 10K BOOT1
RF4 10K BOOT0
Default: Embedded SRAM Mode
J6
BOOT0
BOOT1
MP1
MP2
0533980367
GND BOOT1 to enter System Memory Bootloader Mode
GND BOOT0 to enter User Flash Memory Mode
USB TYPE C FOR DATA TRANSFER
J1
VBUS
CC1
SSRXn2
SSTXn2
SSRXn1
SSTXp2
RFU1
SSTXp1
SSTXn1
Dp1
Dn1
GND
MH1
MH2
MP3
MP4
R1 5.1K
R2 5.1K
MC-318FB
DECOUPLING CAP
VS
C1 220uF
GND
ESD PROTECTION
D14
USB_DM
USB_DP
ESD5302F-3/TR
INPUT CONNECTORS
30VDC
F1
P1
P2
PRT-10474
BOARD TO BOARD CONNECTOR
J3
VCC_5V
VDD_3V3
LED_R
LED_G
USB_DM
SPI_CS
USB_DP
SPI_SCK
CAN_NL
MOSI
CAN_NH
MISO
BOOT0
DRV_STEP
LED_B
DRV_DIR
BUTTON
DIAG0_SWN
TMC_2
DRV_ENN
TMC_1
TEMP_OUT
BOOT1
M55-7012642R
TVS DIODES
D12
D13
PESD2CAN,215
RJ45 FOR CAN BUS
CAN IN
CAN OUT
J2
J5
0855115001
VS
VCC_5V
VDD_3V3
VOUT_5V
GND